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AUTOMATED STUDENTATTENDANCE MONITORING AND ANALYTICS SYSTEM FOR COLLEGE

A PROJECT REPORT

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SPECIALIZATION DIVISION)

BONAFIDE CERTIFICATE

Certified that this report "**AUTOMATED STUDENT ATTENDANCE MONITORING AND ANALYTICS SYSTEM FOR COLLEGE**" is a bonafide work of "J MONICA JASMIINE (20221CSE0030), VAIBHAVI NARIYOOR (20221CSE0025), HARSHITA MALTESH (20221CSE0029)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-26.

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DECLARATION

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ABSTRACT

Traditional attendance management methods in educational institutions often rely on manual roll calls or paper-based records, which consume valuable class time and are prone to human errors and proxy attendance. To address these limitations, this project presents *FaceSync*, an AI-enabled attendance system that automates student attendance using facial recognition technology and a modern backend architecture. The system utilizes computer vision techniques to detect and recognize student faces from uploaded classroom images. During registration, each student's facial features are converted into numerical embeddings and securely stored in a database. During attendance sessions, detected faces are compared with stored embeddings using a similarity-based matching algorithm, enabling accurate identification with confidence scoring, while a session-based tracking mechanism prevents duplicate entries.

The backend is developed using FastAPI, enabling efficient API communication, real-time processing, and seamless integration with frontend applications. Student information and attendance records are maintained using a structured SQLite database for secure storage and fast retrieval. In addition to automated attendance marking, the system offers analytical features such as daily attendance statistics, absentee tracking, identification of students at academic risk, and CSV-based report export for administrative use. By combining artificial intelligence, database management, and RESTful services, the system reduces administrative workload, improves accuracy, and enhances transparency, transforming traditional attendance processes into a reliable, scalable, and data-driven academic management solution.

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CSV	Comma-Separated Values
DB	Database
GPA	Grade Point Average
JSON	JavaScript Object Notation
LIME	Local Interpretable Model-agnostic Explanations
MERN	MongoDB, Express.js, React.js, Node.js
ML	Machine Learning
NLP	Natural Language Processing
REST	Representational State Transfer
RF	Random Forest
SHAP	Shapley Additive Explanations
TF-IDF	Term Frequency – Inverse Document Frequency
UI	User Interface
XAI	Explainable Artificial Intelligence

Chapter 1
INTRODUCTION

Attendance monitoring plays a vital role in educational institutions as it reflects student participation, discipline, and academic engagement. Traditionally, attendance has been recorded through manual roll calls or paper registers, which require significant classroom time and administrative effort. These conventional approaches are often affected by human errors, proxy attendance, and difficulties in maintaining accurate long-term records.

With the rapid advancement of artificial intelligence and computer vision technologies, educational systems are gradually shifting toward automation and digital transformation. Automated attendance systems provide an efficient alternative by minimizing manual intervention while improving reliability and transparency. Facial recognition, in particular, offers a contactless and secure method of identification since every individual possesses unique facial characteristics.

The FaceSync AI-Enabled Attendance System is designed to modernize attendance management using facial recognition and a scalable backend architecture. The system automatically detects and identifies students from images, records attendance in real time, prevents duplicate entries, and generates analytical insights to support academic monitoring. By integrating artificial intelligence with database-driven analytics, the project aims to simplify attendance processes while improving institutional efficiency.

1.1 Background

Educational institutions have long relied on manual attendance recording methods. While these methods are simple to implement, they consume instructional time and often lead to inaccuracies due to manual handling. As class sizes increase and institutions adopt digital learning environments, traditional attendance practices become inefficient and difficult to manage.The emergence of artificial intelligence and image-processing technologies has enabled the development of automated identification systems. Facial recognition systems analyze facial patterns and convert them into numerical representations that can be compared quickly and accurately. Such systems eliminate the need for physical interaction and reduce opportunities for impersonation.

The increasing availability of affordable cameras, high-performance computing libraries, and lightweight backend frameworks has made AI-based attendance solutions practical for realworld academic environments.

1.2 STATISTICS

Studies across educational institutions indicate that manual attendance processes can consume up to 5–10 minutes per lecture, reducing effective teaching time over an academic semester. Human-recorded attendance is also susceptible to recording mistakes and proxy entries, which can compromise academic fairness.

Research in automated biometric systems shows that AI-based facial recognition solutions can achieve recognition accuracy levels above 90–95% under controlled conditions. Automated systems significantly reduce administrative workload and enable real-time monitoring of attendance trends. Furthermore, analytics-driven attendance monitoring helps institutions identify students with low participation early, allowing timely academic intervention.

1.3 PRIOR EXISTING TECHNOLOGIES

Before AI-based attendance systems, several technologies were used to automate attendance tracking:

• Manual Roll Call Systems: Faculty members call student names and record attendance manually. This approach is time-consuming and error-prone.

• RFID-Based Systems: Students carry identification cards scanned by RFID readers. Although faster than manual methods, cards can be shared, enabling proxy attendance.

• Fingerprint Biometric Systems: These provide better identity verification but require physical contact, raising hygiene concerns and causing delays when many students are present.

• Password or Login-Based Systems: Web or mobile-based attendance marking depends on user honesty and lacks strong identity verification.

While these technologies improved automation to some extent, they still faced limitations related to accuracy, convenience, scalability, and security.

1.4 PROPOSED APPROACH

The proposed FaceSync system introduces an AI-driven attendance framework using facial recognition and a modern API-based backend. During student registration, facial images are processed to generate unique embeddings representing facial features. These embeddings are securely stored in a database instead of raw biometric data.

During attendance marking, classroom images are analyzed to detect faces, generate embeddings, and compare them with stored records using similarity measurements. A predefined threshold ensures reliable identification, while a session-tracking mechanism prevents duplicate attendance marking within the same class.

The system also integrates analytics modules that calculate attendance statistics, identify students at academic risk, and generate downloadable reports. The backend is implemented using FastAPI, enabling efficient communication and scalability.

1.5 OBJECTIVES

The primary objectives of the proposed system are:

• **To automate student attendance using facial recognition technology.**

This objective focuses on removing the need for manual attendance by using a system that can recognize students' faces automatically. When a classroom image is uploaded, the system detects and identifies each student and marks their attendance without any human effort, making the process faster and more efficient.

• **To eliminate proxy attendance and manual recording errors.**

Manual attendance methods allow students to mark attendance for others (proxy) and can lead to mistakes like wrong entries or missed names. This system ensures that only the actual student present in the classroom is marked, reducing cheating and improving accuracy.

• **To reduce classroom time spent on attendance procedures.**

Taking attendance manually can take several minutes of valuable teaching time in every class. By automating the process, attendance can be completed within seconds, allowing teachers to spend more time on teaching and learning activities.

• **To maintain structured and secure attendance records.**

The system stores attendance data in an organized digital format, making it easy to access and manage. It also ensures that the data is securely stored in a database, reducing the risk of loss, damage, or unauthorized access compared to paper records

. • **To provide real-time attendance analytics and reporting.**

The system can instantly generate reports such as daily attendance, percentage of attendance, and trends over time. This helps teachers and administrators quickly understand student participation and make informed decisions.

• **To identify students with low attendance for early academic intervention.**

By analyzing attendance records, the system can highlight students who frequently miss classes. This allows teachers to take early action, such as counseling or support, to help those students improve their academic performance.

• **To develop a scalable and user-friendly backend system.**

The system is designed to handle increasing numbers of students and data without performance issues. At the same time, it is built to be easy to use and integrate with other applications, ensuring smooth operation for both developers and users.

1.6 STAINABLE DEVELOPMENT GOALS (SDGS)

The project aligns with several United Nations Sustainable Development Goals:

• **SDG 4 – Quality Education:** Enhances learning efficiency by reducing administrative interruptions and improving academic monitoring.

• **SDG 9 – Industry, Innovation, and Infrastructure:** Promotes technological innovation through AI-driven automation systems.

Fig1.1 Sustainable development goals

1.7 OVERVIEW OF PROJECT REPORT

This project report is organized into multiple chapters to provide a structured understanding of the system:

• **Chapter 1:** Introduction, background, objectives, and project motivation.

• **Chapter 2:** Literature survey and analysis of existing attendance technologies.

• **Chapter 3:** Problem definition and system requirements.

• **Chapter 4:** System architecture and design methodology.

• **Chapter 5:** Implementation details including facial recognition and database integration.

• **Chapter 6:** Results, performance evaluation, and discussion.

• **Chapter 7:** Conclusion and future enhancements.

Chapter 2

LITERATURE REVIEW

The evolution of attendance management systems has been strongly influenced by advancements in IoT technologies, biometric authentication, and computer vision techniques. Early research primarily focused on replacing traditional manual attendance registers with automated digital solutions to improve efficiency and data accuracy.

Initial IoT-based attendance systems relied on fingerprint authentication integrated with web technologies. In one implementation, a fingerprint sensor connected to a Wi-Fi-enabled microcontroller transmitted attendance records directly to a MySQL database hosted on a web server [1]. This approach demonstrated how IoT infrastructure could automate attendance recording while enabling real-time data updates and remote monitoring. The development of attendance management systems has evolved significantly over time, beginning with basic automated solutions and progressing toward more intelligent and integrated systems. Early research focused on IoT-based attendance systems , which introduced automation by connecting devices through the internet. These systems enabled real-time data collection and remote monitoring, improving efficiency compared to traditional manual methods. However, they still relied on hardware components and user interaction, which limited their flexibility in large-scale implementations.

Subsequent studies explored RFID-based attendance tracking, where students were identified through RFID cards scanned by classroom readers [2]. Although this method simplified attendance collection and reduced manual effort, researchers identified significant limitations such as proxy attendance caused by card sharing. These challenges highlighted the need for more secure biometric verification methods. RFID-based attendance systems were introduced to simplify attendance marking by allowing users to scan identification cards. While these systems reduced manual effort and improved speed, they were still prone to issues such as proxy attendance and dependency on physical devices. These limitations highlighted the need for more secure and contactless solutions.

The integration of web technologies introduced centralized attendance platforms capable of data processing and automated report generation [3]. These systems utilized dashboards to compute attendance percentages and present information visually, improving administrative decision-making and academic monitoring. attendance systems were improved to provide

better data handling and structured record management. These systems enabled institutions to store attendance data in an organized manner and generate reports when required, making the process more efficient compared to earlier methods . The introduction of such systems helped reduce errors and improved overall management of attendance records.

Research focusing on institutional deployment emphasized scalability and database integration. University-level automated attendance systems combined identity verification with administrative analytics to ensure reliable large-scale implementation [4]. Databasebacked solutions were shown to improve accuracy compared to traditional paper-based attendance methods.

Earlier desktop-based attendance systems developed using VB.NET and MSSQL improved data storage and retrieval processes [5]. However, these applications required manual login authentication and lacked real-time verification, limiting their suitability for dynamic classroom environments. some systems focused on improving storage and retrieval of attendance data through structured platforms. These systems ensured that attendance information could be accessed easily and maintained properly for future use . However, they still required manual interaction, which limited their effectiveness in real-time scenarios.

A major technological shift occurred with the introduction of surveillance-camera-based attendance systems, which enabled contactless identification through face detection and recognition pipelines [6]. These systems reduced physical interaction, improved hygiene, and minimized classroom disruptions while automating attendance collection. To overcome these limitations, more advanced systems were introduced that aimed to reduce manual dependency and improve automation. These systems provided better efficiency by enabling faster attendance recording and minimizing delays during classroom sessions . As a result, attendance systems became more practical and suitable for regular use in educational environments.

Biometric attendance systems integrating fingerprint sensors with IoT hardware, PHP-based dashboards, and MySQL databases further demonstrated the benefits of combining hardware and software components [7]. In addition to attendance recording, these systems provided graphical analytics that supported performance tracking and trend analysis. Further improvements were made by integrating additional features that enhanced system functionality and performance. These systems not only recorded attendance but also supported better monitoring and management, helping institutions maintain accurate and reliable records . This made attendance systems more useful for both students and administrators.

Recent advancements introduced deep learning-based facial recognition systems capable of real-time identity verification. One study implemented a Convolutional Neural Network combined with the ArcFace algorithm to achieve high recognition accuracy in a university environment [8]. The system also automated attendance compliance by enforcing institutional rules such as minimum attendance requirements. With technological advancements, newer systems focused on improving accuracy and efficiency by using more advanced techniques. These systems were capable of handling larger datasets and performing tasks more effectively, ensuring better performance in different conditions . This marked a significant improvement compared to earlier systems.

Hybrid systems combining facial recognition with speech recognition technologies were also proposed to enhance usability [9]. These solutions enabled automated attendance confirmation through audio interaction, reducing manual classroom interruptions and improving user experience. Moreover, the integration of multiple technologies helped in enhancing the overall system performance. These systems combined different approaches to overcome the limitations of individual methods, resulting in improved accuracy and reduced errors . Such systems provided more reliable and efficient attendance tracking.

More recent research expanded attendance systems beyond simple presence detection by incorporating classroom analytics and focus monitoring [10]. These systems analyzed student engagement alongside attendance data, representing a shift toward intelligent classroom management solutions. Finally, recent developments have focused on improving the overall usability and effectiveness of attendance systems. These systems are designed to be more user-friendly and efficient, ensuring smooth operation and better performance in real-world environments . As a result, attendance systems have become more advanced, reliable, and suitable for modern educational institutions.

SUMMARY OF LITERATURES REVIEWED

Reference	Focus Area	Key Findings	Accuracy / Performance	Limitations
[1]	IoT-based fingerprint attendance	Fingerprint sensor integrated with Wi-Fi module for automated attendance	Real-time data synchronization	Requires physical contact; hygiene concerns
[2]	RFID-based attendance system	RFID cards used for quick student identification	Fast scanning process	Proxy attendance possible; low security
[3]	Web-based attendance system	Centralized dashboard for attendance tracking and reporting	Improved data visualization	Depends on manual input; no automation
[4]	Scalable university system	Database-integrated system for large-scale attendance management	High data reliability	Limited real-time processing; no AI integration
[5]	Desktop-based system (VB.NET & MSSQL)	Efficient data storage and retrieval using local systems	Moderate performance	No real-time tracking; manual authentication
[6]	Camera-based face detection system	Contactless attendance using image processing techniques	Reduced manual effort	Affected by lighting and image quality
[7]	IoT-based biometric system with dashboard	Combines fingerprint authentication with web-based analytics	Real-time monitoring	Still requires physical interaction

[8]	Deep learning face recognition (CNN + ArcFace)	High accuracy facial recognition using deep learning models	High accuracy (>90%)	High computational cost; requires powerful hardware
[9]	Hybrid face and speech recognition system	Multi-modal system improves usability and interaction	Improved user experience	Complex implementation; affected by noise
[10]	Intelligent attendance with analytics	Attendance system with engagement monitoring and analytics	Advanced analytics and insights	Privacy concerns; high infrastructure requirements

Table 2.1 Summary of Literature reviews

Research Gaps

- Dependence on manual input or physical interaction
- Prone to proxy attendance and human errors
- Hardware dependency increases cost and reduces flexibility
- Lack of real-time automation
- Affected by environmental factors (lighting, camera angle, quality)
- Requires continuous user involvement
- Some systems need high computational resources
- Complex implementation in advanced systems
- Limited scalability for large institutions
- Not fully efficient or reliable in real-world conditions

To address these gaps, the proposed project implements a CNN with ResNet-based face recognition system that is fully automated, contactless, and more reliable in real-time environments. It reduces manual effort, prevents proxy attendance, improves accuracy under varying conditions, and provides a scalable and efficient solution for modern attendance management.

Chapter 3
METHODOLOGY

The proposed FaceSync attendance system follows a structured workflow that combines image processing, facial recognition, database management, and analytical evaluation to automate attendance marking. The methodology is designed to ensure accuracy, reliability, and efficient real-time processing while preventing duplicate entries and maintaining organized records. The overall workflow consists of image validation, face recognition, similarity matching, attendance verification, database storage, and analytics generation.

A. Image Capture and Input Validation

The attendance process begins when a faculty member uploads a classroom image for attendance marking or submits a student photograph during registration. The system first verifies whether the uploaded file is valid by checking image format compatibility and ensuring successful decoding.

If the image cannot be processed or no face is detected, the system immediately returns a structured error response to avoid incorrect processing. During student registration, the system strictly allows only one face in the image to prevent enrollment errors. In contrast, classroom attendance images may contain multiple students, and each detected face is processed independently.

Additionally, important metadata such as timestamp and class identifier are attached to every request. This ensures proper tracking, traceability, and organized storage of attendance records.

B. Face Detection and Embedding Extraction

After validation, the system performs face detection using a computer vision pipeline. The uploaded image is converted into a suitable color format to ensure compatibility with the recognition engine.

Each detected face is transformed into a numerical representation called a **facial embedding**, which captures unique facial characteristics in vector form. The embedding can be represented mathematically as:

$$E=[e1,e2,e3,\ldots,e128]$$

Where:

- E represents the facial embedding vector
- ei represents individual facial feature values
- 128 denotes the embedding dimension

This conversion transforms visual facial information into structured numerical data, enabling

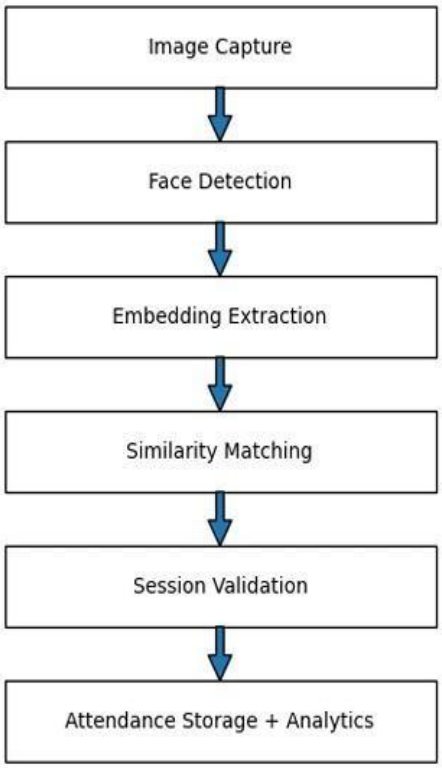

Fig 3.1 Methodology Workflow

C. Similarity Matching Using Euclidean Distance

Once embeddings are generated, the system retrieves all previously stored student embeddings from the database. Identity verification is performed by calculating the Euclidean distance between the detected embedding Ed and stored embedding Es.

The similarity distance is calculated as:

$$D(E_d, E_s) = \sqrt{\left\{\sum_{\{i=1\}}^{\{n\}} \left(E_{\{di\}} - E_{\{si\}}\right)^2\right\}}$$

Where:

- D represents similarity distance
- n=128 embedding dimensions
- Smaller distance values indicate higher similarity

A match is confirmed only when: $D(E_d, E_s) < T$ where Tis a predefined threshold (typically 0.6).

The confidence level of recognition is computed as:

$$Confidence = 1 - D(E_d, E_s)$$

This mechanism ensures attendance is recorded only when recognition reliability meets acceptable accuracy limits.

D. Session Validation and Duplicate Prevention

Before storing attendance, the system verifies whether the identified student has already been marked present during the same class session. A unique session key is generated using the current date combined with the class identifier.

If attendance has already been recorded for that session, the system returns an **"already marked"** response. Otherwise, a new attendance entry is created containing student details, timestamp, confidence score, and class ID. This process prevents duplicate attendance records and maintains data consistency.

E. Attendance Analytics and Risk Identification

After attendance recording, analytical metrics are automatically updated. The system calculates each student's attendance percentage using:

$$\text{Attendance Percentage} = \frac{\text{Classes Attended}}{Total\ classes} \times 100$$

Students whose attendance falls below 75% are categorized as **"At Risk."** This feature enables administrators and faculty members to identify students requiring early academic intervention and monitoring.

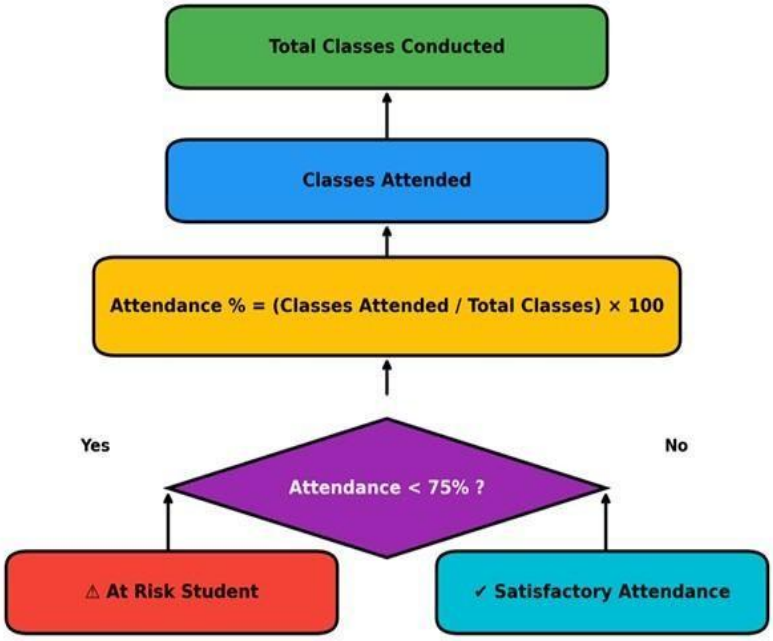

Fig 3.2 Attendance Calculation Flow

F. Database Modeling and Structured Storage

The system employs a relational database design to maintain clear separation between student identity information and attendance records. Each student is assigned a universally unique identifier (UUID), ensuring accurate tracking and eliminating duplication.

The **Student table** stores personal details such as name, roll number, registration timestamp, photo reference, and serialized facial embeddings. The **Attendance table** records attendance events, including timestamp, confidence score, and class identifier.

This structured schema allows efficient querying for date-wise, class-wise, and student-wise attendance analysis while avoiding redundant data storage.

G. Embedding Optimization and Retrieval Strategy

To improve performance, facial embeddings are stored in JSON format and converted into numerical arrays only during comparison. Instead of repeatedly querying embeddings

for every face, the system retrieves all embeddings once per attendance session and performs vectorized comparisons.

This optimization reduces database overhead and improves processing speed. The matching process follows linear time complexity:

$$\text{Time Complexity}=O(n)$$

where nnn represents the number of registered students. This ensures predictable performance even as the database grows.

H. Threshold Calibration Strategy

The matching threshold significantly affects system accuracy. A lower threshold reduces false acceptance but may reject valid matches, whereas a higher threshold increases flexibility but risks incorrect identification.

Experimental testing is conducted using multiple classroom samples to determine an optimal threshold value. Performance is evaluated using:

$$FAR = \frac{\text{False Acceptances}}{\text{Total Impostor Attempts}}$$

$$FRR = \frac{\text{False Rejections}}{\text{Total Genuine Attempts}}$$

Threshold calibration minimizes both False Acceptance Rate (FAR) and False Rejection Rate (FRR), ensuring balanced performance in real classroom environments.

I. API Communication and Response Handling

The backend system is implemented using FastAPI, which manages communication through RESTful API endpoints. These endpoints handle student registration, attendance marking, analytics retrieval, and report export operations.

Each API response contains structured information including:

- Status indicator

- Student details (if recognized)
- Confidence score
- Error message (if applicable)

This standardized communication enables smooth integration with frontend applications while improving maintainability and scalability.

FACENET-BASED CNN MODEL WITH RESNET ARCHITECTURE

A. Model Overview

The proposed attendance system is built using a deep learning-based face recognition model known as the FaceNet-based Convolutional Neural Network (CNN) with a ResNet architecture. This model is specifically designed to extract meaningful facial features and represent them in a compact numerical form. The primary objective of the model is to convert facial images into embeddings that can uniquely identify individuals in a reliable and efficient manner.

Unlike traditional machine learning models such as Support Vector Machines or Random Forest classifiers, the system does not perform direct classification of identities. Instead, it follows a metric learning approach, where the model learns to measure the similarity between faces. This allows the system to determine whether two faces belong to the same person based on the distance between their embeddings. This approach improves scalability and allows the system to handle large numbers of students without retraining the model frequently.

Each detected face is transformed into a 128-dimensional embedding vector that captures essential facial characteristics. These embeddings are stored in the database and used for realtime comparison during attendance marking. This design ensures high accuracy, fast processing, and efficient identity verification in classroom environments.

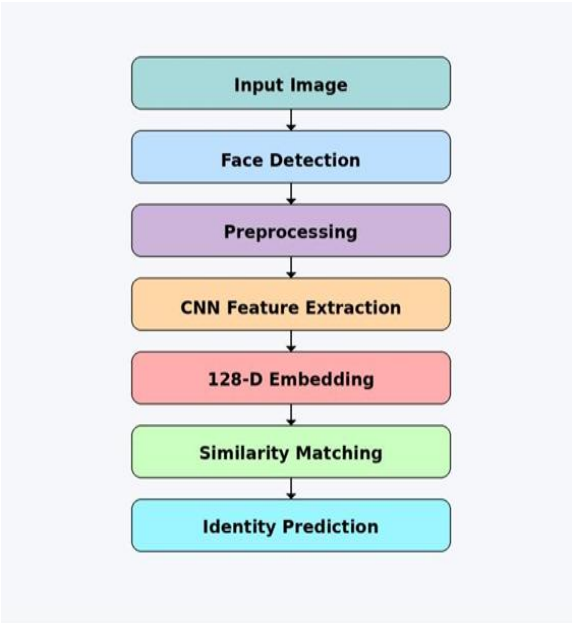

Fig 3.3 ML Model Architecture Diagram

B. Model Architecture and Layers

The architecture of the model consists of multiple layers that work together to extract and refine facial features. The process begins with the input layer, which accepts a preprocessed facial image resized to a standard size such as 160×160 pixels in RGB format. Preprocessing ensures uniformity across all input images and improves recognition consistency.

The convolutional layers play a key role in identifying important visual features such as edges, contours, and textures. These layers progressively learn complex patterns that represent different facial structures. Pooling layers are used to reduce the dimensionality of the feature maps while preserving important information. This helps in improving computational efficiency and reducing redundancy in the data.

After feature extraction, the data passes through fully connected layers, which convert the extracted features into a compact representation. The final layer generates a 128-dimensional embedding vector that acts as a unique identity representation. Instead of using a classification layer, the system compares embeddings using Euclidean distance to determine similarity, ensuring accurate and flexible recognition.

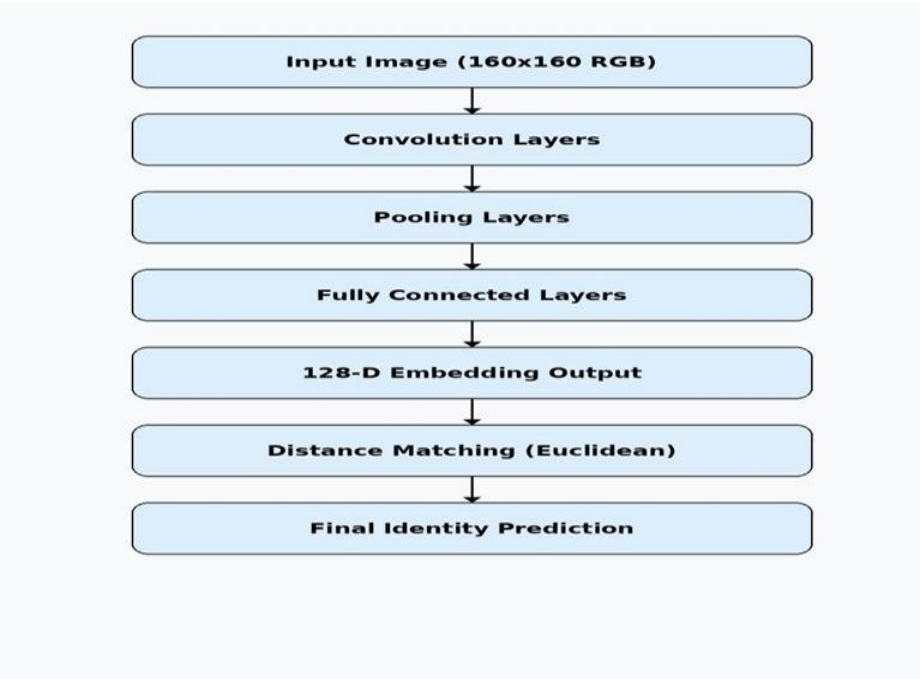

Fig 3.4 FaceNet-Based CNN Model Architecture

C. Dataset Description

The performance of the face recognition model depends heavily on the quality and diversity of the training dataset. To achieve robust performance, the system utilizes publicly available datasets such as VGGFace2 and Labeled Faces in the Wild (LFW). VGGFace2 contains millions of images with variations in pose, lighting, age, and facial expressions, enabling the model to learn generalized facial features.

LFW is a widely used benchmark dataset that contains images captured in real-world conditions. It is particularly useful for evaluating the model's performance under unconstrained environments. These datasets together provide a strong foundation for training and validating the model, ensuring that it performs well in practical scenarios.

In addition to public datasets, the system also uses a custom dataset collected during student registration. Each student provides a facial image, which is processed to generate embeddings. These embeddings are stored in the database and used during attendance marking. This combination of public and custom datasets ensures both generalization and domain-specific accuracy.

Dataset path- attendence-main/backend/attendance.db

D. Training Pipeline

The training pipeline consists of several stages that ensure the model learns effective facial representations. The first stage involves data collection, where images are gathered from public datasets and student registration inputs. These images are then preprocessed to ensure consistency in size, format, and quality.

The next stage involves face detection, where the system identifies and extracts facial regions from the input images. Once the faces are detected, feature extraction is performed using the CNN model, which captures important facial characteristics. These features are then converted into 128dimensional embeddings that represent each individual.

The model is trained using a metric learning approach with triplet loss, which helps in learning discriminative features. During training, the model is evaluated using performance metrics such as accuracy and precision. This pipeline ensures that the model is capable of handling real-world variations and provides reliable recognition performance.

E. Triplet Loss Function

The model uses a specialized loss function known as triplet loss to learn facial similarity effectively. Unlike traditional classification-based loss functions, triplet loss focuses on learning the relative distance between embeddings. This allows the model to differentiate between similar and dissimilar faces more effectively.

A triplet consists of three images: an anchor image, a positive image belonging to the same individual, and a negative image belonging to a different individual. The model is trained to minimize the distance between the anchor and positive images while maximizing the distance between the anchor and negative images. This ensures that embeddings of the same person are closer together in the feature space.

The use of triplet loss enables the model to create a well-structured embedding space, where identities are clearly separated. This improves recognition accuracy and makes the system more robust to variations in lighting, pose, and facial expressions. As a result, the system can reliably identify students in real-time attendance scenarios.

F. V Model

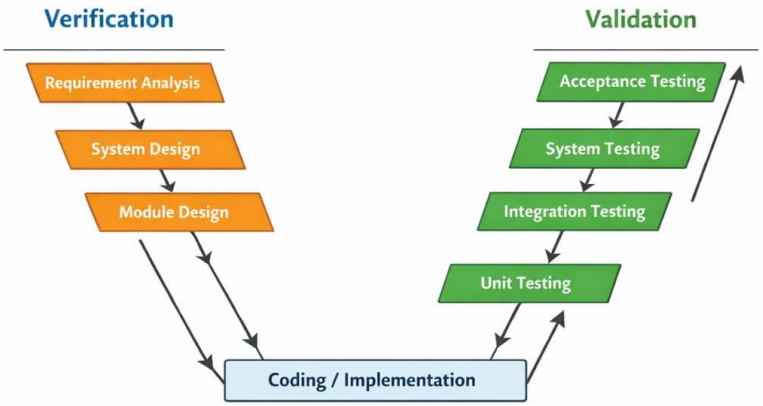

Fig 3.5 V Model

The V-Model (Verification and Validation Model) is used in this project to ensure that every stage of development is properly tested. It follows a structured approach where each development phase has a corresponding testing phase, helping to improve accuracy and reliability of the system.

Application of V-Model

(Development / Verification)

Requirement Analysis:

Identify the need for an automated attendance system to replace manual methods and reduce proxy attendance.

System Design:

Design the overall system architecture including modules like student registration, face detection, recognition, and database.

Module Design:

Design individual components such as CNN + ResNet model, face detection algorithm, and database structure.

Implementation (Coding):

Develop the system using tools like Python, OpenCV, and integrate the trained face recognition model.

(Testing / Validation)

Unit Testing:

Test individual modules like face detection and feature extraction.

Integration Testing:

Check if modules (camera, model, database) work together correctly.

System Testing:

Test the complete system to ensure proper attendance marking.

Acceptance Testing:

Validate the system with users to confirm it meets project requirements.

Chapter 4

PROJECT MANAGEMENT

Effective project management ensured the timely completion and successful integration of the complex machine learning and web development components. This chapter outlines the project timeline, risk analysis.

The project timeline was managed using a Gantt Chart, a visual, bar-chart-based project management tool that tracks tasks, mile stones, and deadlines. This was crucial for coordinating the parallel works teams of data preparation, model development, and web application coding. (Show at autarkical representation for object break down, project' schedule, out lining tasks, milestones, and deadlines in chronological order)

Major Task	W1	W2	W3	W4	W5	W6	W7	W8
Project Initiation	X							
Selection of Topic	X							
Literature Review (v)		X	X					
Design and Analysis			X	X	X	X	X	
System Requirement Phase (vi)			X	X				
System design phase (vii)				X	X			
Functional unit design phase (viii)					X	X	X	
Report							X	X
Proposal					X			

Table 4.1 Project planning timeline

Description: Table 4.1 summarises the timeline during the initial project planning phase. The projectbeganwiththeselectionofthetopicandproceededtooverlaptheliteraturereviewand the design and analysis phases, which culminated in the proposal. The detailed functional unit design phase overlapped with the start of implementation planning in later weeks. The milestones are identified and the deadlines are... structured weekly to ensure progress on both foundational research (Literature Review) and technical blueprint creation (Design and Analysis).

Week 1 (W1): Project Initiation and Topic Selection

During the first week, the project was initiated with basic planning and understanding of objectives. The group discussed various ideas and finalized the project topic based on feasibility and relevance.

Week 2 (W2): Literature Review (Initial Phase)

In this week, the group began reviewing existing research papers and attendance systems. This helped in understanding current technologies and identifying limitations in existing methods.

Week 3 (W3): Literature Review and Initial Design

The literature review was continued, and initial design and analysis of the system were started. Key system requirements were identified based on the study of existing solutions.

Week 4 (W4): System Requirement and Design Phase

During this week, the system requirements were finalized, and the design phase was carried forward. The overall system architecture and workflow were planned.

Week 5 (W5): System Design and Proposal Submission

The detailed system design, including modules and functionalities, was developed. The project proposal was also prepared and submitted during this phase.

Week 6 (W6): Functional Unit Design Phase

In this week, the focus was on designing individual functional units of the system, ensuring proper interaction between different components.

Week 7 (W7): Final Design and Report Preparation

The functional design was completed, and report writing was initiated. Documentation of system design, methodology, and analysis was carried out.

Week 8 (W8): Report Completion and Submission

In the final week, the report was completed, reviewed, and refined. Final corrections were made, and the project documentation was prepared for submission.

4.2 Risk Analysis

A PESTLE analysis (Political, Economical, Societal, Technological, Legal, Environmental) wasconductedtoassesshowthesefactorsmightimpacttheproject'ssuccessandtoformulate proactive risk mitigation strategies.

Technological risks were particularly significant, including potential incompatibility between Python libraries, cloud deployment challenges, and memory limitations when loading multiple large ML models simultaneously. The project mitigated these risks through modular design, containerized environments (optional), and efficient memory handling using lazy loading techniques.

Factor	Description of Risk	Mitigation Strategy
P Political	Changes in government or policies could affect data accessibility (CGWB/NAQUIM data).	Early data acquisition and local secure storage of the initial dataset.
E Economical	Unexpected costs for cloud hosting required for ensemble model training.	Utilize cost-effective, open-source ML libraries (XGBoost/CatBoost) and budget cloud environments.
S Societal	User distrust in AI predictions over traditional, human-led methods (lack of adoption).	Transparent model explanations and rigorous public validation/demonstration of superior accuracy.
T Technological	Inaccuracy of ML models, poor data quality/sparsity in certain regions.	Implement high-performing ensemble/stacking models to minimize prediction error; use robust data imputation techniques.
L Legal	Compliance with India's Digital Personal Data Protection Act (DPDPA) concerning the handling and display of user location data.	Ensure the system does not store or process personally identifiable information (PII) beyond the session.
E Environmental	Long-term geological changes (e.g., extreme monsoons) may gradually degrade the model's predictive accuracy over time.	Plan for periodic model retraining with the most recent CGWB/geospatial data (e.g., on a yearly cycle).

Table 4.2 PESTEL analysis

4.3 Gant Chart

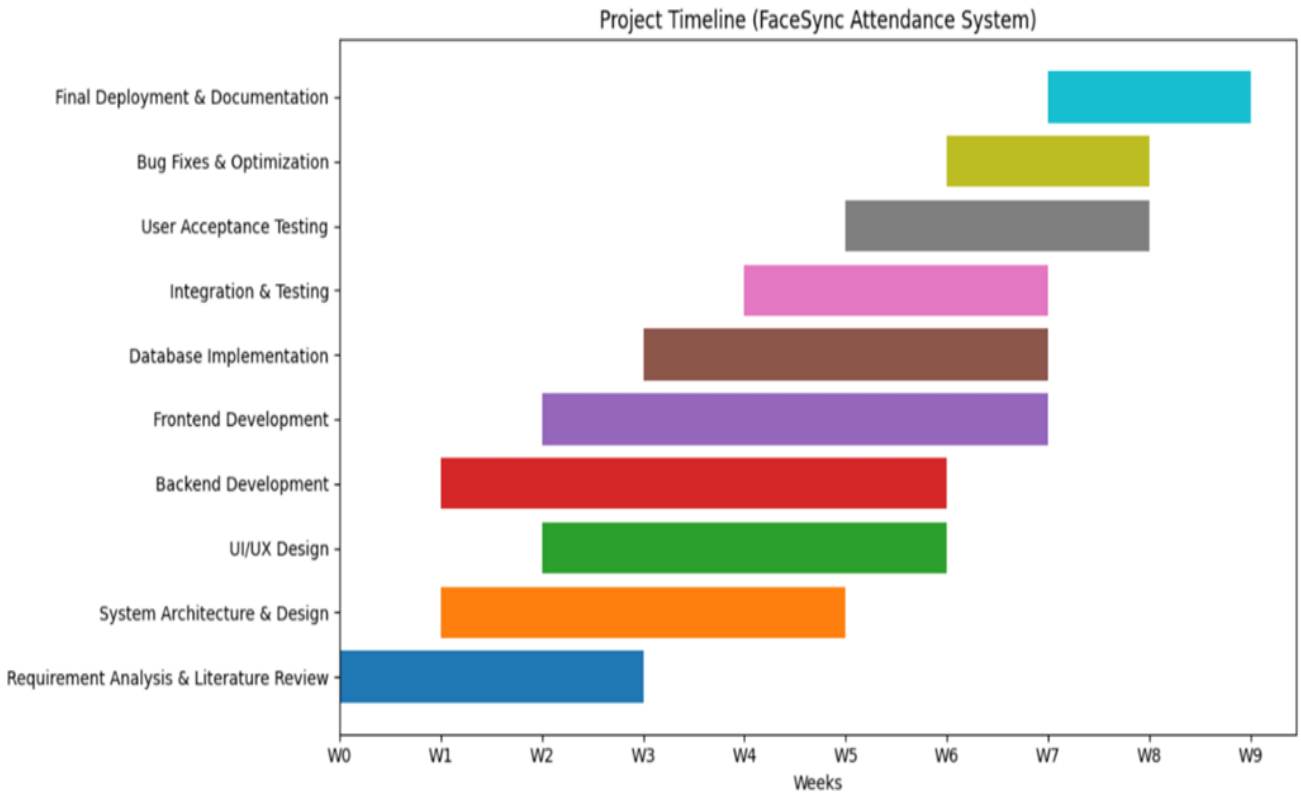

Fig 4.1 Gant chart

Chapter 5

ANALYSIS AND DESIGN

The Analysis and Design phase converts the conceptual idea of the FaceSync AI-Enabled Attendance System into a structured technical solution. This phase identifies system requirements, defines architecture, models data relationships, and explains operational workflows. The design focuses on automation, reliability, scalability, and efficient interaction between artificial intelligence modules and database services.

The proposed system eliminates manual attendance processes by integrating facial recognition, RESTful APIs, and structured database storage. The design ensures that attendance is recorded accurately while maintaining performance efficiency even when the number of registered students increases

5.1 Requirements

System requirements describe the expected behavior and performance characteristics necessary for successful implementation.

5.1.1 Functional Requirements

The system must support the following operations:

• Student registration using a facial image The system allows students to register by uploading a facial image, which serves as their unique identity within the platform. This image is processed and stored securely, forming the foundation for future recognition and attendance tracking.

• Automatic face detection from uploaded images Once an image is uploaded, the system automatically detects the presence and location of faces within it using computer vision techniques, ensuring that only valid facial data is processed further.

• Extraction of facial embeddings for identity representation The detected face is converted into a numerical representation called a facial embedding, which captures distinctive facial features in a compact form for efficient comparison and identification.

• Comparison of detected faces with stored student data The system compares the extracted facial embeddings with those stored in the database to determine if the face matches any registered student, enabling accurate identification.

• Automatic attendance marking upon successful recognition When a student is successfully recognized, the system automatically marks their attendance without requiring manual input, improving efficiency and reducing human error.

• Prevention of duplicate attendance entries during the same session To maintain data integrity, the system ensures that a student's attendance is recorded only once per session, preventing duplicate entries even if the face is detected multiple times.

• Storage of attendance records with timestamp and confidence value. Each attendance entry is stored along with the exact time it was recorded and a confidence score indicating the reliability of the facial recognition match.

• Retrieval of attendance data based on date and class The system provides functionality to fetch attendance records based on specific dates and class identifiers, making it easy to review and manage data.

• Generation of attendance statistics and analytics It processes stored data to generate meaningful insights such as attendance percentages, trends, and patterns, helping educators make informed decisions.

• Identification of students with attendance below required limits The system automatically identifies students whose attendance falls below predefined thresholds, enabling timely intervention and monitoring.

• Exporting attendance data into CSV format Users can export attendance records into CSV files, allowing easy sharing, reporting, and integration with other tools like spreadsheets.

• Providing API endpoints for frontend interaction The system exposes API endpoints that enable seamless communication with frontend applications, allowing features like registration, attendance marking, and data retrieval to be accessed through user interfaces.

5.1.2 Non-Functional Requirements Accuracy

Accuracy:

ensures that the system correctly identifies students without false matches or missed detections. It uses facial embeddings and optimized similarity thresholds to balance false acceptance and rejection rates. High-quality images and preprocessing techniques further improve recognition performance.

Proper threshold tuning and continuous testing help maintain accuracy under different conditions. The system remains consistent as the database grows by using efficient embedding comparisons and filtering invalid inputs, making it reliable in varied environments.

Performance:

Performance ensures that attendance is marked quickly in real time without delays. The system processes images, detects faces, and matches them efficiently, even when handling multiple students at once.

Optimized libraries like OpenCV and NumPy improve processing speed, while FastAPI enables asynchronous request handling. Reusing embeddings and minimizing database queries further enhance system efficiency on standard hardware.

Scalability:

Scalability allows the system to handle increasing numbers of students and records without slowing down. Efficient database design and structured storage help maintain performance as data grows.

Embedding-based matching enables fast comparisons even with large datasets. The modular architecture also supports future expansion, including cloud integration and multi-classroom support.

Security:

Security protects sensitive biometric data by storing facial embeddings instead of raw images. This ensures privacy and reduces the risk of misuse.

Access control and API validation prevent unauthorized access, while secure data handling and backups protect system integrity. These measures ensure safe and reliable data management.

Usability:

Usability ensures the system is simple and easy for faculty to use. Users can register or mark attendance by uploading images, requiring minimal technical knowledge.

Clear feedback messages and automation reduce manual effort and confusion. The system can also integrate with web or mobile interfaces for flexible access.

Maintainability:

Maintainability allows the system to be easily updated and fixed. Its modular design ensures that issues can be resolved without affecting the entire system.

Structured coding and frameworks like FastAPI improve organization. This makes future enhancements and feature additions easier to implement.

Reliability:

Reliability ensures consistent system performance without failures. Error-handling mechanisms prevent crashes and handle invalid inputs effectively.

Validation checks at multiple stages ensure correct data processing. Backup and logging features further maintain data integrity and dependable operation over time

5.2 Block Diagram

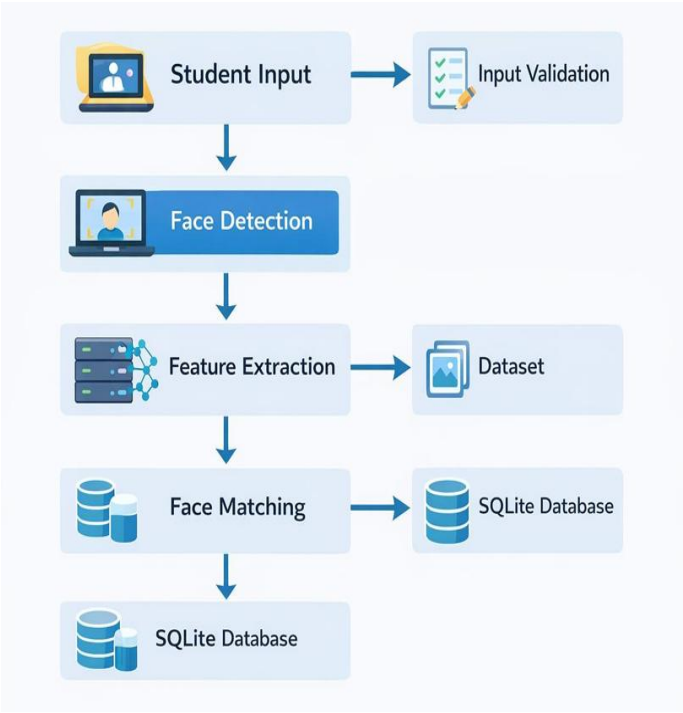

Fig 5.1 Block Diagram

The block diagram illustrates how different components of the system interact to perform attendance automation.

System Components

1. **Image Input Module**

The Image Input Module serves as the entry point of the system, responsible for accepting images during both student registration and attendance marking processes. It supports image uploads from various sources such as webcams, mobile cameras, or stored files, ensuring flexibility in real-world usage. The module ensures that images are received in supported formats and prepares them for further processing. By handling both single-face images for registration and multi-face images for classroom attendance, the module adapts to different system requirements efficiently.

This module plays a crucial role in maintaining the quality of input data, as the accuracy of the entire system depends heavily on the clarity and correctness of the images provided. It ensures that images are properly decoded and converted into a format suitable for processing by computer vision algorithms. Proper handling of image resolution and format consistency helps improve downstream tasks such as face detection and feature extraction.

Additionally, the module attaches metadata such as timestamps and session identifiers to each input image. This information is essential for tracking attendance sessions and maintaining structured records. By ensuring reliable input handling and preparation, the Image Input Module forms the foundation for accurate and efficient system operation

2. Validation Module

The Validation Module ensures that the uploaded images meet the necessary criteria before further processing. It verifies the image format, checks for decoding errors, and ensures that the file is not corrupted. This step is essential to prevent invalid inputs from affecting the system's performance or causing unexpected failures during processing.

During student registration, the module enforces strict validation rules by allowing only images containing a single face. This prevents incorrect enrollment and ensures that each student is uniquely represented in the database. For attendance images, the module allows multiple faces but ensures that at least one valid face is detected before proceeding further.

The module also handles error reporting by providing clear feedback when validation fails. For example, if no face is detected or the image format is unsupported, the system returns an appropriate error message. This improves user experience and ensures that only valid data enters the system pipeline, enhancing reliability and robustness.

3. Face Detection Module

The Face Detection Module is responsible for identifying and locating human faces within the uploaded images. It uses computer vision techniques to scan the image and detect facial regions accurately. This module plays a critical role in separating relevant facial data from the background, enabling further processing steps.

The module is capable of detecting multiple faces in a single image, which is essential for classroom attendance scenarios. Each detected face is extracted as an individual region of interest, ensuring that all students present in the image are processed independently. This improves system efficiency and ensures that no valid face is missed during attendance marking.

To ensure robustness, the module performs well under varying lighting conditions, facial orientations, and partial occlusions. By accurately detecting faces across different scenarios, the system maintains high reliability. The effectiveness of this module directly impacts the accuracy of the overall recognition system.

4.Embedding Extraction Module

The Embedding Extraction Module converts detected facial images into numerical feature vectors known as embeddings. These embeddings represent unique facial characteristics and serve as the foundation for identity recognition. By transforming visual data into structured numerical form, the system enables efficient comparison between individuals.

This module uses a deep learning-based CNN model to extract high-level features from facial images. These features capture important details such as facial geometry, texture, and structure. The resulting 128-dimensional embedding vector provides a compact yet highly descriptive representation of each face.

The use of embeddings improves both accuracy and efficiency, as numerical comparisons are faster than image-based matching. Additionally, embeddings are more secure than storing raw

images, as they cannot be easily reversed into original images. This module is a key component in enabling reliable and scalable face recognition.

5. Face Matching Engine

The Face Matching Engine compares newly generated embeddings with stored embeddings in the database to determine identity. It uses Euclidean distance to measure similarity between feature vectors, where smaller distances indicate higher similarity.

The engine applies a predefined threshold to decide whether a match is valid. If the computed distance is below the threshold, the identity is confirmed; otherwise, it is rejected. This approach ensures accurate recognition while minimizing false matches. Threshold calibration plays a key role in balancing system performance.

The matching process is optimized to handle multiple comparisons efficiently. By retrieving embeddings once per session and performing vectorized operations, the system reduces processing time. This ensures that the engine can operate in real time, even in large classroom environments.

6. Attendance Processing Module

The Attendance Processing Module is responsible for recording attendance after successful identity verification. Once a student is recognized, the system creates an attendance entry with relevant details such as timestamp, class identifier, and confidence score.

To ensure data integrity, the module includes a duplicate prevention mechanism. It checks whether the student has already been marked present in the same session. If a duplicate entry is detected, the system prevents re-recording and returns an appropriate response.

This module ensures that attendance records are accurate, consistent, and free from duplication. By automating attendance marking, it significantly reduces manual effort and improves overall efficiency in classroom management.

7. Database System

The Database System stores all essential data, including student profiles and attendance records. It uses a structured relational design to maintain clear separation between different types of data. Each student is assigned a unique identifier to ensure accurate tracking.

Student information includes details such as name, roll number, and facial embeddings, while attendance records include timestamps and session identifiers. This structured storage allows efficient querying and retrieval of data for analysis and reporting.

The database is optimized for performance and scalability, ensuring that it can handle large datasets without degradation. It also supports secure storage practices, protecting sensitive biometric data from unauthorized access.

8. Analytics Engine

The Analytics Engine processes attendance data to generate meaningful insights. It calculates attendance percentages for each student based on recorded sessions and identifies patterns over time.

The system applies predefined thresholds to identify students with low attendance. These students are categorized as at-risk, enabling early intervention by faculty. This feature supports proactive academic management.Additionally, the engine generates visual summaries and reports, transforming raw data into actionable insights. This enhances decision-making and improves transparency in attendance monitoring.

9. API Layer

The API Layer acts as a bridge between the frontend interface and backend processing system. It handles communication through structured HTTP requests and responses, ensuring seamless interaction.

Each API endpoint is designed to perform specific tasks such as registration, attendance marking, and analytics retrieval. The use of FastAPI ensures high performance and efficient request handling.

The API layer also ensures data consistency and security by validating inputs and managing responses. This modular communication structure improves system scalability and maintainability.

The modular design allows each block to operate independently while contributing to overall system functionality.

5.3 System Flow Chart

The system workflow follows a sequential decision-based process ensuring accurate recognition and storage.

Flow Steps

1. User uploads image.
2. System validates image format.
3. Face detection is performed.
4. Facial embeddings are generated.
5. Stored embeddings are retrieved from database.
6. Similarity comparison is executed.
7. Threshold condition is checked.
8. If matched:
   - o Verify session duplication.
   - o Store attendance record.
9. If not matched:
   - o Mark as unknown.
10. Update analytics.
11. Send structured response to user.

This flow ensures efficient processing while preventing incorrect attendance entries.

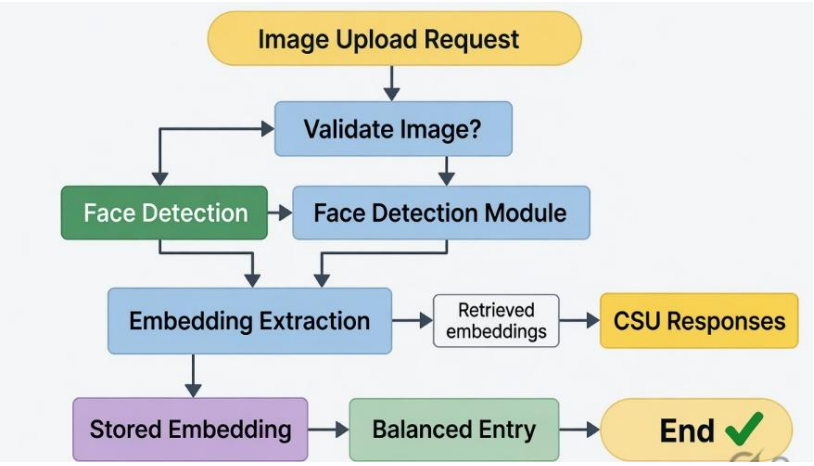

Fig 5.2 System Flow Chart

5.4 Choosing Devices

The system is designed to minimize hardware dependency while maintaining efficiency.

Hardware Selection

• Standard webcam or mobile camera for image capture.

• Faculty computer or smartphone for uploading attendance images.

• Server or local machine for backend execution.

Software Selection

• Fast API for backend services.

• OpenCV for image handling.

• Face recognition library for biometric processing.

• SQLite database for lightweight storage.

• NumPy for mathematical operations.

These choices ensure cost-effective deployment suitable for educational institutions.

Fig 5.3 Choosing Devices

5.5 Designing units

The system is divided into logical functional units.

Registration Unit:

The Registration Unit is responsible for enrolling students into the system by capturing and processing their facial data. During registration, the system accepts a single facial image of the student, ensuring that the input meets the required standards for accurate recognition. The module verifies that only one face is present in the image, preventing incorrect enrollment and ensuring that each student is uniquely represented within the database. This step is critical for maintaining data integrity and avoiding ambiguity during later recognition processes.

Once the image is validated, the system proceeds to generate facial embeddings using the feature extraction module. These embeddings represent the unique characteristics of a student's face in numerical form. Instead of storing raw images, the system securely stores these embeddings along with essential student details such as name, roll number, and registration timestamp. This approach enhances both efficiency and privacy, as embeddings are lightweight and less vulnerable to misuse compared to raw image data.

The Registration Unit also ensures that duplicate entries are avoided by checking existing records before adding new data. If a student is already registered, the system can prevent redundant storage or allow controlled updates. This unit forms the foundation of the entire system, as the accuracy of recognition depends heavily on the quality and correctness of registered data.

Recognition Unit:

The Recognition Unit is responsible for processing classroom images and identifying multiple faces present within them. Unlike the registration phase, where only a single face is allowed, this unit is designed to handle group images containing several students. It detects all visible faces in the uploaded image and prepares them for further processing by isolating each face as an individual entity.

Once faces are detected, the unit ensures that each extracted face is properly aligned and preprocessed before passing it to the embedding extraction stage. This preprocessing step is essential to maintain consistency across different images, especially when dealing with variations in lighting, angle, and facial expressions. By standardizing the input, the system improves recognition accuracy and reduces potential errors.

The Recognition Unit operates efficiently even in complex classroom environments where multiple faces may overlap or appear at different scales. Its ability to detect and isolate faces accurately ensures that no student is missed during attendance marking. This unit plays a vital role in enabling real-time, contactless attendance processing.

Matching Unit:

The Matching Unit is responsible for comparing the embeddings generated from detected faces with the stored embeddings in the database. It uses a similarity measurement technique, typically Euclidean distance, to determine how closely two embeddings match. A smaller distance indicates a higher likelihood that the faces belong to the same individual.

To ensure reliable identification, the unit applies a predefined threshold value. If the computed distance falls below this threshold, the system confirms the identity of the student. Otherwise, the face is classified as unknown. This threshold-based approach helps balance accuracy and flexibility, reducing both false positives and false negatives.

The Matching Unit is optimized for performance by using efficient data retrieval and comparison techniques. Instead of repeatedly querying the database, embeddings are loaded once and compared using vectorized operations. This ensures that the system can handle multiple comparisons quickly, making it suitable for real-time attendance scenarios in large classrooms.

Attendance Management Unit:

The Attendance Management Unit is responsible for recording attendance once a student has been successfully recognized. It creates structured attendance entries containing essential information such as student identity, timestamp, class identifier, and confidence score. This ensures that attendance data is stored in a consistent and organized manner.

One of the key features of this unit is duplicate prevention. The system checks whether a student has already been marked present during the same session using a session-based tracking mechanism. If a duplicate attempt is detected, the system prevents additional entries and maintains data integrity. This ensures that attendance records remain accurate and trustworthy.

The unit also supports efficient data retrieval for future use, enabling easy access to attendance logs for analysis and reporting. By automating the attendance recording process, it significantly reduces manual effort and eliminates common errors associated with traditional methods.

Analytics Unit:

The Analytics Unit transforms raw attendance data into meaningful insights that support academic decision-making. It processes attendance records to calculate key metrics such as attendance percentage, total classes attended, and absentee trends for each student. These metrics provide a clear understanding of student participation over time.

The system uses predefined thresholds, such as 75% attendance, to identify students who may be at academic risk. These students are flagged automatically, allowing faculty members to take timely corrective actions. This proactive approach helps improve student performance and ensures better academic outcomes.

In addition to individual analysis, the unit can generate class-level insights and trends, enabling institutions to monitor overall attendance patterns. By converting raw data into actionable information, the Analytics Unit enhances transparency and supports data-driven decision making in educational environments.

Export Unit:

The Export Unit provides functionality for generating and downloading attendance reports in a structured format. It allows users to export attendance data as CSV files, making it easy to analyze and share information using external tools such as spreadsheets. This feature is particularly useful for administrative tasks and record-keeping.

The unit ensures that exported data is well-organized and includes relevant details such as student names, attendance status, timestamps, and calculated percentages. This structured output simplifies reporting and helps institutions maintain accurate records for compliance and evaluation purposes.

Additionally, the Export Unit supports quick and efficient data retrieval, ensuring that reports can be generated on demand without delays. By providing accessible and portable data formats, this unit enhances the usability of the system and supports better academic management practices.

5.6 Standards

The system follows established software and data standards:

• RESTful API communication principles.

• JSON format for request and response data.

• ISO date-time format for timestamps.

• UUID identifiers for unique record management.

• Database normalization techniques.

• Structured error handling practices.

• Secure storage of biometric information.

These standards ensure interoperability and long-term system sustainability.

5.7 Domain Model Specification

The domain model defines system entities and relationships.

Student Entity Attributes:

• Student ID (UUID)

• Name

• Roll Number

• Facial Embedding

• Photo Path

• Registration Timestamp **Attendance Entity** Attributes:

• Attendance ID

• Student ID

• Student Name

• Roll Number

• Timestamp

• Confidence Score

• Class Identifier

Relationship

One student can have multiple attendance records, while each attendance record belongs to one student.This relational structure enables efficient filtering and reporting.

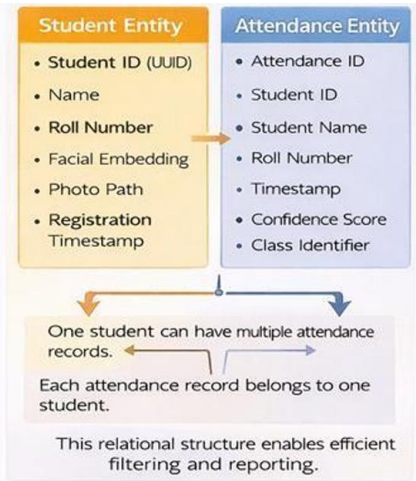

Fig 5.4 Functional Requirements

5.8 Communication Model

The system uses a client–server architecture. Communication Process

1. Client sends HTTP request through frontend interface.

2. Fast API backend receives and validates request.

3. Face recognition processing is executed.

4. Database operations are performed.

5. JSON response returned to client. Key API Endpoints

• Student registration API.

• Attendance marking API.

• Attendance statistics API.

• Attendance export API.

This model ensures flexible integration with web or mobile applications.

5.9 Functional View

The functional view represents system behavior from the user's perspective.

Primary Functions

• Register students.

• Recognize faces automatically.

• Mark attendance instantly.

• Prevent duplicate entries.

• Generate analytics reports.

• Identify at-risk students.

• Export attendance data.

Each function contributes to reducing administrative workload and improving efficiency.

5.10 Operational View

The operational view explains real-time system behavior during classroom usage.

1. Faculty uploads classroom image.

2. Backend processes image immediately.

3. Faces are detected and recognized simultaneously.

4. Attendance entries are stored automatically.

5. Session tracking prevents repeated marking.

6. Analytics update dynamically.

7. Reports become instantly available for export.

The system operates continuously with predictable performance due to optimized embedding comparison and database design.

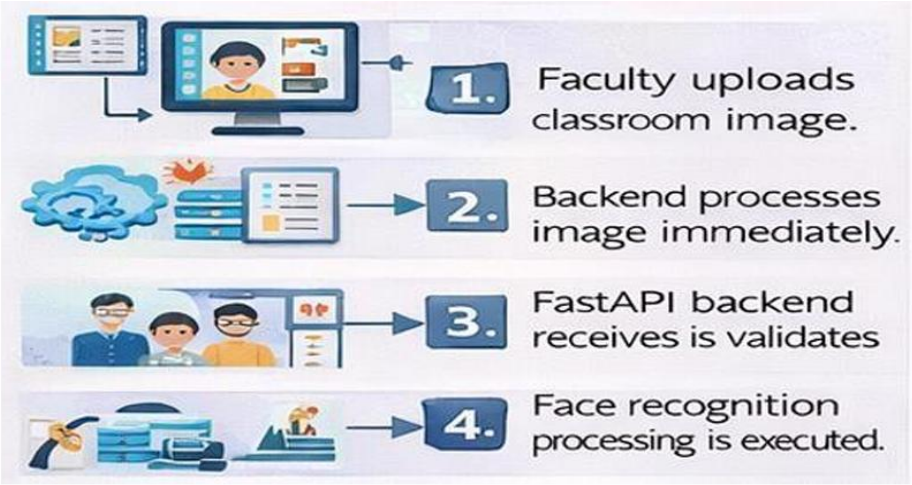

Fig 5.5 Operational View

Chapter 6

H ARDWARE, SOFTWARE AND SIMULATION

This chapter explains the technological environment used to implement the FaceSync AIEnabled Attendance System. The project primarily focuses on software-based intelligent automation; therefore, hardware requirements are minimal while software components play a major role in system functionality. The chapter describes hardware configuration, development tools, program structure, and simulation/testing procedures used to validate system performance.

6.1 Hardware

Unlike traditional biometric attendance systems that require dedicated fingerprint scanners or RFID readers, the proposed system is designed to operate using commonly available computing devices. This approach reduces implementation cost while maintaining high operational efficiency.

Hardware Components Used

1. **Image Capture Device**

A standard digital camera, laptop webcam, or mobile phone camera is used to capture student images.

These images serve two purposes:

• Student registration (single portrait image)

• Classroom attendance marking (group image)

The system processes uploaded images rather than relying on specialized biometric sensors.

2. Processing System

A personal computer or laptop acts as the processing unit where the backend server runs. The system performs computational tasks such as:

• Face detection

• Embedding extraction

• Similarity comparison

• Database operations

Recommended configuration:

• Processor: Intel i5 / Ryzen 5 or higher

• RAM: Minimum 8 GB

• Storage: 10 GB free space

• Operating System: Windows/Linux/macOS

3. Server Environment

The Fast API backend runs locally or on a server machine. The server handles API requests, executes recognition algorithms, and manages database communication.

4. Network Connectivity

Internet or local network connectivity enables communication between frontend applications and backend APIs through HTTP requests.

6.2 Software Development Tools

The FaceSync system is developed using modern open-source software tools that support artificial intelligence, web services, and database management. Programming Language

Python

Selected due to strong AI and computer vision ecosystem.

o Provides extensive libraries for image processing and machine learning.

Backend Framework

• Fast API

o web framework. High-performance

o Supports asynchronous request handling.

o Automatically generates API documentation.

o Enables structured RESTful communication.

Computer Vision Libraries

• **OpenCV (cv2)**

o preprocessing. Image decoding and

o Color format conversion.

• **face_recognition**

o Face detection and embedding generation.

o facial comparison. Mathematical Processing Distance-based

• **NumPy**

o Efficient numerical operations.

o Vector-based embedding comparisons.

Database Technology

• **SQLite**

o Lightweight relational database.

o Suitable for academic and small-scale deployments.

• **SQLAlchemy ORM**

o Object-relational mapping between Python classes and database tables.

Additional Tools

• Uvicorn – ASGI server to run FastAPI applications.

• Pillow (PIL) – Image handling support.

• VS Code – Development environment.

• Postman / Browser – API testing.

6.3 Software Code

The software implementation consists of two major components:

1. Database Model Layer

2. Backend API and Recognition Engine

Database Model Design

The database uses SQL Alchemy ORM to define structured entities.

Student Table

Stores registered student information including:

• Unique UUID identifier

• Name and roll number

• Serialized facial embedding

• Stored image path

• Registration timestamp

Facial embeddings are stored as JSON data and converted back into numerical arrays during recognition. This approach allows efficient storage while maintaining compatibility with numerical comparison algorithms.

Attendance Table

Records attendance events containing:

• Student identity

• Timestamp

• Confidence score

• Class identifier

Each attendance record references a student, forming a one-to-many relationship.

Backend API Implementation

The FastAPI backend provides multiple endpoints:

Student Registration API

• Accepts student details and image.

• Detects face and extracts embedding.

• Stores embedding and metadata in database.

• Ensures only one face is registered per student.

Attendance Marking API

• Accepts classroom image.

• Detects multiple faces.

• Compares embeddings using distance calculation.

• Marks attendance if similarity threshold is satisfied.

• Prevents duplicate marking using session tracking.

Analytics APIs Provide:

• Attendance statistics

• Present/absent counts

• At-risk student identification

• CSV export functionality

Recognition Algorithm

Each detected face is converted into a 128-dimension embedding vector. Matching is performed by calculating distance between embeddings. Lower distance indicates higher similarity, ensuring reliable identity verification.

SOFTWARE CODE

INSTALL REQUIREMENTS

```
pip install fastapi uvicorn face_recognition opencv-python numpy sqlalchemy
```

DATABASE (models.py)

```
 from sqlalchemy import Column, String, DateTime
from sqlalchemy.ext.declarative import declarative_base
import uuid
 from datetime import datetime
Base = declarative_base()
class Student(Base):
__tablename__ = "students
" id = Column(String, primary_key=True,
 default=lambda: str(uuid.uuid4()))
 name = Column(String) Column(String) # stored as string
class Attendance(Base):
 __tablename__ = "attendance"
 id = Column(String, primary_key=True, default=lambda: str(uuid.uuid4()))
 student_id = Column(String)
 timestamp = Column(DateTime, default=datetime.utcnow)
```

DATABASE CONNECTION (database.py)

```
 from sqlalchemy import create_engine
from sqlalchemy.orm import sessionmaker
from models import Base
DATABASE_URL="sqlite:///./attendance.db"
```

```
engine= create_engine(DATABASE_URL)

SessionLocal = sessionmaker(bind=engine)

Base.metadata.create_all(bind=engine)

def get_db():

 db = SessionLocal()

 try:

 yield db

finally:

 db.close()
```

FACE PROCESSING (face_service.py)

```
 import face_recognition

import numpy as np

def get_embedding(image_path):

 image = face_recognition.load_image_file(image_path)

 encodings = face_recognition.face_encodings(image)

if len(encodings) == 0: return None return encodings[0] def compare_faces(known_embeddings, new_embedding, threshold=0.5):

distances = face_recognition.face_distance(known_embeddings, new_embedding) if len(distances) == 0:

 return None min_index =

np.argmin(distances) if

distances[min_index] < threshold:

return min_index

return None
```

FASTAPI BACKEND (main.py)

```
 from fastapi import FastAPI, UploadFile, File, Depends

 from sqlalchemy.orm import Session import shutil

 import numpy as np
```

```
from database import get_db
from models import Student, Attendance
from face_service import get_embedding, compare_faces
app = FastAPI()
# REGISTER STUDENT
@app.post("/register")
def register(name: str, file: UploadFile = File(...), db:Session = Depends(get_db)):
path = f"temp_{file.filename}"
with open(path, "wb") as buffer:
shutil.copyfileobj(file.file, buffer)
embedding = get_embedding(path)
if embedding is None:
return{"error": "No face detected"}
student = Student
( name=name,
embedding=str(embedding.tolist())
```

db.add(student) db.commit() return {"message":

"Student registered successfully"}

ATTENDANCE API

```
@app.post("/mark-attendance") def mark_attendance(file: UploadFile = File(...),
db: Session = Depends(get_db)):
path = f"temp_{file.filename}"
with open(path, "wb") as buffer:
shutil.copyfileobj(file.file, buffer)
new_embedding =
get_embedding(path) if
new_embedding is
```

```
None: return

{"error": "No face detected"}

students = db.query(Student).all()

known_embeddings = []

for s in students:

 emb = np.array(eval(s.embedding))

known_embeddings.append(emb) index =

 compare_faces(known_embeddings, new_embedding)

return {"message": "Unknown person"}

student = students[index]

# Check duplicate attendance existing

= db.query(Attendance).filter( Attendance.student_id =

= student.id ).first()

if existing: r

eturn {"message": "Attendance already marked"}

 attendance = Attendance(student_id=student.id)

 db.add(attendance) db.commit() return {"message": f"{student.name}
```

Chapter 7

EVALUATION AND RESULT

The evaluation phase of the FaceSync AI-Enabled Attendance System was carried out to measure the performance, accuracy, and reliability of the developed solution under practical operating conditions. The objective of this phase was to verify whether the system successfully automates attendance marking using facial recognition while maintaining correctness, efficiency, and stability. Multiple testing scenarios were designed to simulate real classroom environments, including student registration, attendance marking using group images, database updates, and analytics generation.

7.1 Test Points

The system was evaluated across several functional areas to ensure that every module operated as expected. The student registration process was tested to confirm that only valid images containing a single face were accepted and that duplicate entries were prevented. Face detection capability was examined using images with different lighting conditions and multiple individuals to verify consistent detection performance. Recognition accuracy was assessed by comparing stored facial embeddings with faces captured during attendance sessions, ensuring that registered students were correctly identified while unknown individuals were rejected.

Attendance marking functionality was also tested to ensure automatic recording once a face was successfully recognized. Special attention was given to duplicate attendance prevention, where repeated submissions of the same student image within a session were checked to confirm that attendance was recorded only once. Additionally, database operations were verified to ensure accurate storage, retrieval, and timestamp management of attendance records. The analytics module was tested to validate attendance percentage calculations and report generation features. Error-handling mechanisms were also examined by providing invalid inputs such as empty files and unsupported formats.

7.2 Test Case

Test cases are structured scenarios designed to validate whether each component of the attendance monitoring system functions correctly. They serve as a systematic approach to verify that the system meets its functional and non-functional requirements. Each test case includes predefined inputs, execution steps, and expected outputs, allowing testers to compare actual system behavior with expected outcomes.

The testing process focuses on key modules such as student registration, face detection, recognition, attendance marking, database operations, and analytics. By executing these test cases, the system's accuracy, reliability, and performance are evaluated under different conditions. This ensures that the system performs consistently in real-time classroom environments.

Each test case is documented with essential details such as test case ID, description, preconditions, steps, test data, expected result, actual result, and status. This structured format ensures repeatability, clarity, and easy tracking of system performance during testing.

TC_01 – Student Registration (Valid Input)

- **Description:** Verify successful student registration using a valid facial image.
- **Preconditions:** System is running and database is connected.
- **Test Steps:**
  1. Upload a clear image containing a single face
  2. Enter student details
  3. Submit registration
- **Test Data:** Valid single-face image
- **Expected Result:** Student details and embedding are stored successfully
- **Actual Result:** Student registered and embedding stored correctly

  Status: Pass

TC_02 – Student Registration (Invalid Input)

- **Description:** Verify system behavior when image contains no face.
- **Preconditions:** System is active **Test Steps:**
  1. Upload blank or invalid image

- **Test Data:** Image without face
- **Expected Result:** Error message displayed
  - **Actual Result:** —No face detected error shown
  - **Status:** Pass

TC_03 – Face Detection

- **Description:** Verify detection of faces in classroom image.
- **Preconditions:** Group image available
- **Test Steps:**

  1. Upload classroom image

- **Test Data:** Image with multiple faces
- **Expected Result:** All faces detected
- **Actual Result:** Faces detected accurately
- **Status:** Pass

TC_04 – Face Recognition (Known Person)

- **Description:** Verify recognition of registered student.
- **Preconditions:** Student is already registered
- **Test Steps:**

  1. Upload attendance image

- **Test Data:** Image containing registered student
- **Expected Result:** Student identified correctly
- **Actual Result:** Student recognized successfully
- **Status:** Pass

TC_05 – Face Recognition (Unknown Person)

- **Description:** Verify system behavior for unregistered person.
- **Preconditions:** Unknown face provided
- **Test Steps:**

  1. Upload image

- **Test Data:** Image of unregistered individual
- **Expected Result:** Person marked as ―Unknown‖
- **Actual Result:** Unknown detected correctly
- **Status:** Pass

TC_06 – Attendance Marking

- **Description:** Verify attendance is recorded after recognition.
- **Preconditions:** Student is recognized
- **Test Steps:**
    1. Detect and match face
    2. Trigger attendance recording
- **Test Data:** Registered student image
- **Expected Result:** Attendance recorded with timestamp
- **Actual Result:** Attendance successfully recorded
- **Status:** Pass

TC_07 – Duplicate Attendance Prevention

- **Description:** Ensure duplicate attendance is not recorded.
- **Preconditions:** Student already marked present
- **Test Steps:**
    1. Upload same student image again
- **Test Data:** Same student image
- **Expected Result:** Duplicate entry blocked
- **Actual Result:** Duplicate prevented successfully
- **Status:** Pass

TC_08 – Database Storage

- **Description:** Verify data storage in database.
- **Preconditions:** Attendance marked
- **Test Steps:**
    1. Check database records
- **Test Data:** Attendance entries
- **Expected Result:** Correct data stored
- **Actual Result:** Data stored accurately
- **Status:** Pass

TC_09 – Analytics Calculation

- **Description:** Verify attendance percentage calculation.
- **Preconditions:** Attendance data exists
- **Test Steps:**
    1. Open analytics dashboard
- **Test Data:** Attendance records
- **Expected Result:** Correct percentage displayed
- **Actual Result:** Percentage calculated correctly
- **Status:** Pass

TC_10 – Risk Identification

- **Description:** Identify students with low attendance.
- **Preconditions:** Attendance below threshold
- **Test Steps**
    1. View analytics
- **Test Data:** Low attendance data
- **Expected Result**: Student flagged as “At Risk”
- **Actual Result:**
- **Status:** Pass

TC_11 – API Response Validation

- **Description:** Verify API responses for requests.
- **Preconditions:** Backend running
- **Test Steps:**
  1. Send API request
- **Test Data:** Valid request
- **Expected Result:** Proper JSON response
- **Actual Result:** Correct response received
- **Status:** Pass

7.3 Test Result

The results obtained from testing indicated that the system performed effectively across all major functions. The registration module successfully stored student details and generated facial embeddings without errors. Face detection operated reliably in both individual and group images, enabling accurate identification of registered students. Recognition results showed consistent matching performance, with the system correctly distinguishing between registered and unregistered individuals.

Attendance recording worked automatically once recognition was completed, significantly reducing manual effort. The duplicate prevention mechanism functioned correctly by blocking repeated attendance entries during the same session. Database operations remained stable, with accurate storage of attendance logs and timestamps. The analytics module generated meaningful summaries, including attendance percentages and absentee lists, which could be exported as CSV reports. Error-handling tests confirmed that the system responded with appropriate messages without causing application failure.

Sl. No	Module Tested	Test Description	Expected Outcome	Actual Result	Status
1	Registration Module	Store student details and generate facial embeddings	Student data saved successfully with embedding creation	Student information stored correctly and embeddings generated without errors	Passed
2	Face Detection	Detect faces from individual and group images	System detects all visible faces accurately	Faces detected reliably in both single and multiple-person images	Passed

3	Face Recognition	Match detected faces with registered database	Correct identification of registered students and rejection of unknown users	Accurate matching achieved with consistent performance	Passed
4	Attendance Recording	Automatically mark attendance after recognition	Attendance recorded automatically with timestamp	Attendance marked successfully without manual intervention	Passed
5	Duplicate Prevention	Prevent multiple attendance entries in same session	System blocks repeated attendance entries	Duplicate entries successfully prevented	Passed
6	Database Operations	Store attendance logs and timestamps	Stable and accurate data storage	Records stored correctly with proper timestamps	Passed
7	Analytics Module	Generate attendance summaries and absentee lists	Accurate statistics and report generation	Meaningful analytics generated and displayed correctly	Passed
8	CSV Export	Export attendance reports	Downloadable CSV file generated	CSV reports exported successfully	Passed
9	Error Handling	Handle invalid inputs and system errors	Proper error messages without system crash	System responded safely with appropriate messages	Passed

Table 7.1 Comparative Test Results

7.4 Insights

The evaluation process provided valuable insights into the effectiveness of integrating artificial intelligence into attendance management. One major observation was the significant reduction in time required for attendance recording compared to traditional manual methods. The system demonstrated that automation can improve both efficiency and accuracy simultaneously. It was also observed that recognition performance depends largely on the quality of registration images, emphasizing the importance of proper enrollment procedures.

Another important insight was the effectiveness of session-based duplicate prevention, which enhanced attendance authenticity and minimized misuse. The lightweight architecture allowed smooth operation even on standard hardware, indicating that the system can be deployed in institutions without advanced infrastructure. Furthermore, the

analytics features transformed attendance data into useful academic insights, enabling educators to monitor participation trends and identify students at risk due to low attendance.

Overall, the evaluation confirmed that the proposed system is reliable, efficient, and suitable for real-world educational environments. The results demonstrate that AI-driven automation can successfully modernize traditional attendance systems while improving transparency and data management.

Chapter 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

8.1. Social Aspects

The automated face-recognition attendance system contributes positively to educational and organizational environments by reducing manual workload and improving efficiency. It minimizes time spent on traditional roll-call methods, allowing instructors to focus more on teaching activities. The system promotes transparency because attendance records are generated automatically, reducing favoritism or human error. Additionally, accurate attendance monitoring helps institutions identify irregular participation early and provide support to students when needed. However, user awareness and acceptance are important, as individuals may initially feel uncomfortable with biometric technologies; therefore, proper communication about system usage is necessary.

8.2. Legal Aspects

Since the system collects and processes facial data, it must comply with applicable data protection and privacy regulations. Institutions deploying the system should obtain informed consent from users before capturing facial images. Personal data must be securely stored, accessed only by authorized personnel, and used strictly for attendance purposes. Data retention policies should also be defined, ensuring that biometric information is deleted when no longer required. Following legal guidelines helps prevent misuse of personal information and protects both users and organizations from legal disputes.

8.3. Ethical Aspects

Ethically, the system must respect individual privacy and maintain fairness in recognition performance. Developers should ensure that facial recognition models are trained using diverse datasets to avoid bias toward particular groups. The system should operate transparently, clearly informing users about how their data is collected, processed, and stored. Ethical implementation also requires avoiding surveillance misuse and ensuring that

biometric data is not shared or repurposed without permission. Responsible handling of

sensitive data builds trust among users and promotes ethical technology adoption.

8.4. Sustainability Aspects

The solution supports sustainability by reducing paper usage associated with manual attendance registers and printed reports. Digital record management lowers administrative resource consumption and promotes environmentally friendly practices. Efficient software design and optimized processing reduce computational overhead, helping minimize energy consumption when deployed on institutional systems. Cloud or local database optimization can further support long-term sustainable operation by improving storage efficiency and reducing unnecessary data duplication.

8.5. Safety Aspects

From a safety perspective, the system ensures secure handling of biometric information through encryption, authentication mechanisms, and controlled database access. Proper error handling prevents application crashes and protects stored records from corruption. Backup mechanisms and secure login systems safeguard attendance data against unauthorized access or accidental loss. Additionally, the system reduces physical contact during attendance processes, supporting hygienic and safe environments, especially in shared or crowded spaces.

Chapter 9

CONCLUSION

The developed FaceSync attendance system demonstrates how artificial intelligence can effectively improve attendance management in academic environments. By integrating facial recognition technology, it replaces manual methods with an automated process that saves time and reduces administrative effort. The use of facial embeddings ensures accurate and consistent identity verification, while similarity-based recognition combined with session-level duplicate prevention maintains the authenticity of attendance records. Additionally, the structured database design supports reliable storage and easy retrieval of student and attendance data.

Beyond automation, the system enhances academic monitoring through analytics that provide insights such as attendance percentages and absentee reports. These features help educators make informed decisions and support early intervention for students with low attendance. The modular architecture further ensures scalability and maintainability, allowing the system to grow and adapt without affecting performance. Overall, the project proves that AI-based attendance systems are practical, efficient, and valuable solutions for modern educational institutions, promoting transparency, accuracy, and accountability.

FUTURE WORK

While the current system performs accurately and reliably, several improvements can enhance its capabilities. Integrating real-time video processing would allow continuous attendance marking without manual image uploads. Adding liveness detection can further strengthen security by preventing spoofing using photos or screens. Role-based authentication and secure access mechanisms can also be introduced to provide customized dashboards for different users.

Scalability can be improved by adopting advanced databases like PostgreSQL and deploying the system on cloud infrastructure to support larger datasets and multiple classrooms. Additionally, incorporating predictive analytics using machine learning can help identify attendance trends and at-risk students early. These enhancements would evolve FaceSync into a more advanced and comprehensive academic monitoring system.

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Appendix

A. System Requirements:

Python, FastAPI, OpenCV, NumPy, SQLite, and a system with minimum 4GB RAM and a camera.

B. API Endpoints:

- Register student
- Mark attendance
- View attendance
- Generate report **C. Database:**
- Student (ID, name, class, embedding)
- Attendance (ID, student ID, date, time, confidence)

D. Output:

Attendance logs, recognition results, and CSV reports.

E. Limitations:

Depends on image quality and lighting conditions.

1. Datasets

The dataset plays a crucial role in determining the performance and reliability of the proposed face recognition-based attendance system. A well-structured dataset enables the model to learn meaningful facial features and generalize effectively across different individuals and conditions. In this system, a combination of publicly available datasets and a custom dataset is utilized to achieve both robustness and practical applicability. The use of diverse data sources ensures that the model performs accurately in real-world classroom environments.

Public datasets such as VGGFace2 are used to provide a strong foundation for training the deep learning model. This dataset contains millions of facial images collected from thousands of individuals, covering a wide range of variations including pose, lighting, age, and expressions. Such diversity helps the model learn generalized facial representations, making it more capable of handling variations encountered in real-world scenarios. The large scale of the dataset also contributes to improved feature extraction and recognition accuracy.

Another important dataset used is Labeled Faces in the Wild (LFW), which is widely recognized for evaluating face recognition systems. LFW consists of images captured in unconstrained environments, meaning the faces vary significantly in terms of background, lighting, and orientation. This makes it particularly useful for testing the robustness of the model under realistic conditions. By incorporating LFW, the system ensures that its performance is validated against challenging realworld data.

In addition to public datasets, a custom dataset is created specifically for the attendance system during the student registration phase. Each student provides a facial image, which is processed and converted into a numerical embedding. These embeddings are stored in the database and used for identity verification during attendance marking. This custom dataset ensures that the system is tailored to the specific group of users, improving recognition accuracy within the institution.

The dataset undergoes preprocessing before being used in the model. This includes resizing images to a standard dimension, normalizing pixel values, and ensuring consistent color formats. Preprocessing helps reduce noise and variability in the data, allowing the model to focus on relevant facial features.

It also improves computational efficiency and ensures uniformity across all inputs, which is essential for maintaining consistent model performance. Data diversity and quality are key factors that influence the effectiveness of the system. By combining large-scale public datasets with a domain-specific custom dataset, the system achieves a balance between generalization and specialization. The public datasets help the model learn universal facial features, while the custom dataset ensures accurate recognition of registered students. This hybrid approach enhances both performance and reliability.

Overall, the dataset used in the proposed system is carefully designed to support accurate, scalable, and real-time face recognition. It enables the model to perform well under different environmental conditions and ensures that the system remains effective as the number of users increases. The combination of robust training data and structured deployment data makes the system suitable for practical implementation in educational institutions.

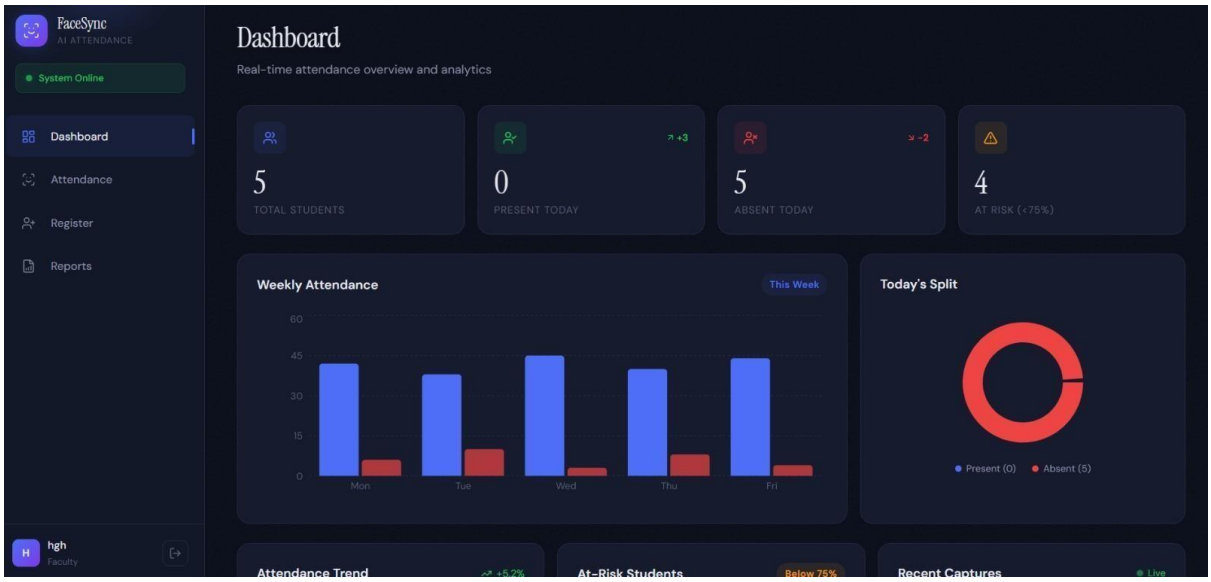
Link dataset- https://www.kaggle.com/code/yhuan95/face-recognition-with-facenet/log

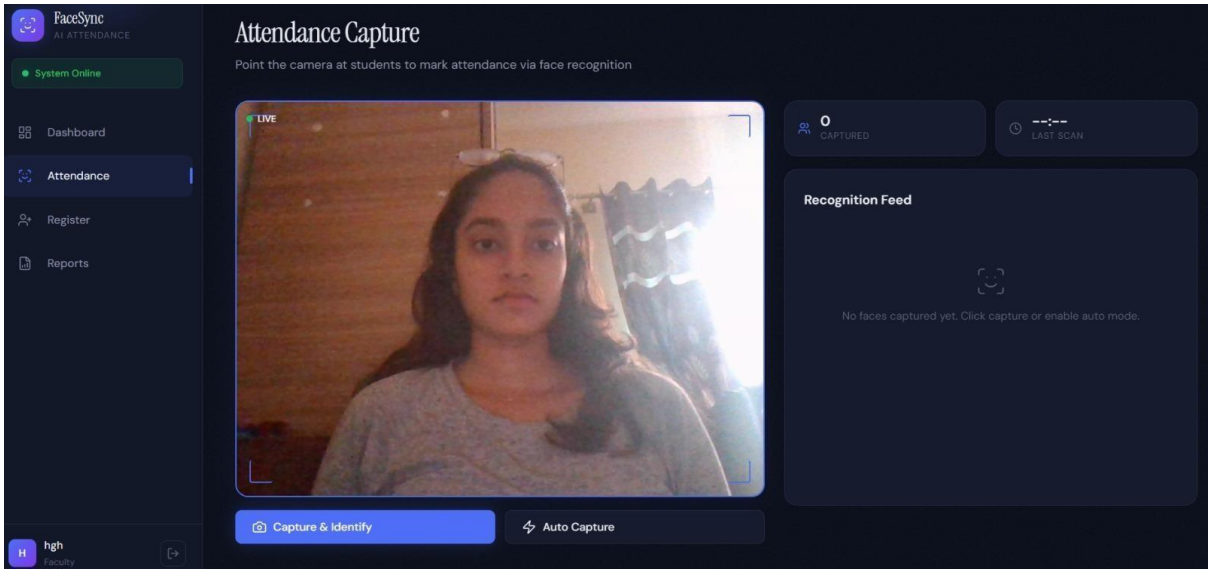
Dataset path- attendence-main/backend/attendance.db

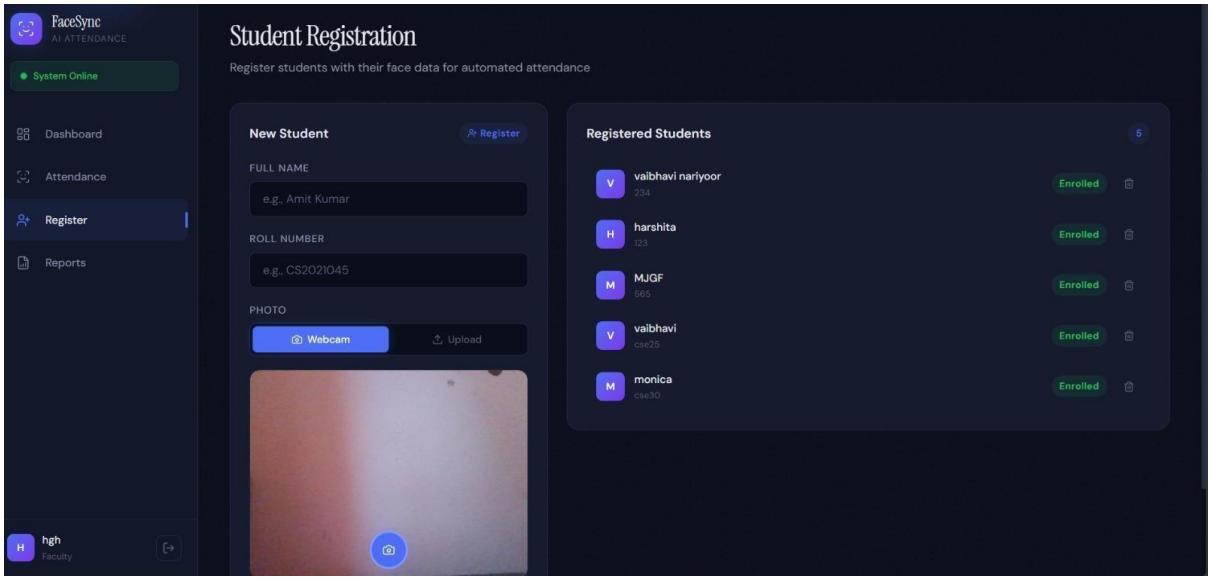
2. Live Project Demo

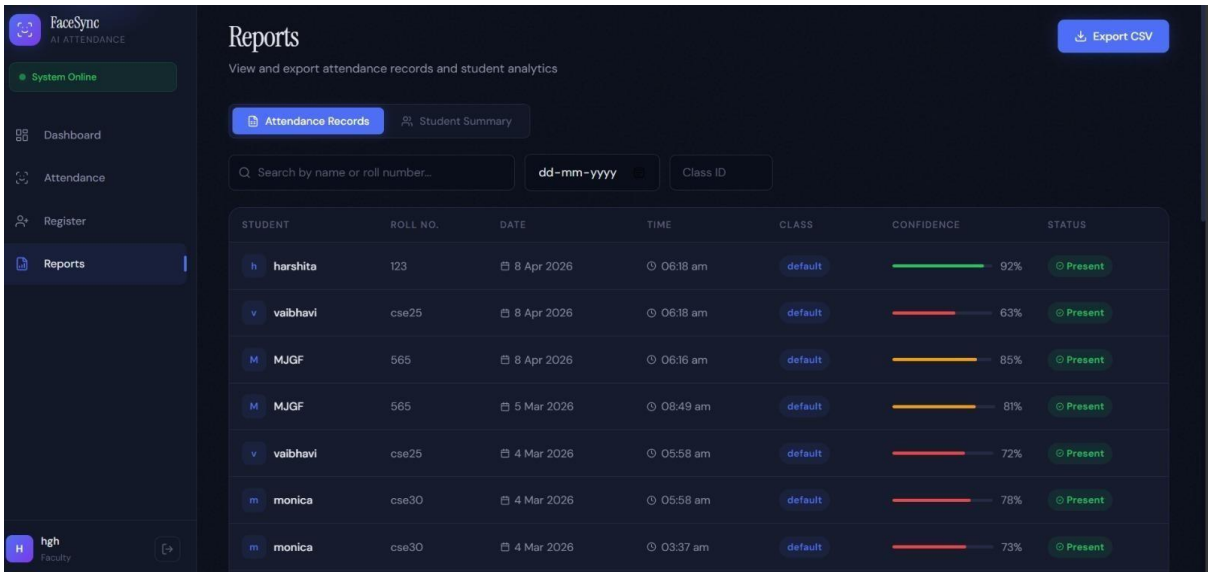
- Github- https://github.com/vnariyoor-netizen/attendence http://localhost:5173/dashboard
- Live Demo- http://localhost:5173/dashboard

3. Screenshots of projects




FaceSync
AI ATTENDANCE
System Online
Dashboard
Attendance
Register
Reports
hgh
Faculty
Attendance Capture
Point the camera at students to mark attendance via face recognition
LIVE
0
CAPTURED
--:--
LAST SCAN
Recognition Feed
No faces captured yet. Click capture or enable auto mode.
Capture & Identify
Auto Capture


FaceSync
AI ATTENDANCE
System Online
Dashboard
Attendance
Register
Reports
hgh
Faculty
Student Registration
Register students with their face data for automated attendance
New Student
Register
FULL NAME
e.g., Amit Kumar
ROLL NUMBER
e.g., CS2021045
PHOTO
Webcam
Upload
Registered Students
5
vaibhavi nariyoor
234
Enrolled
harshita
123
Enrolled
MJGF
565
Enrolled
vaibhavi
cse25
Enrolled
monica
cse30
Enrolled


FaceSync
AI ATTENDANCE
System Online
Dashboard
Attendance
Register
Reports
hgh
Faculty
Reports
View and export attendance records and student analytics
Export CSV
Attendance Records
Student Summary
Search by name or roll number...
dd-mm-yyyy
Class ID
STUDENT
ROLL NO.
DATE
TIME
CLASS
CONFIDENCE
STATUS
harshita 123 8 Apr 2026 06:18 am default 92% Present
vaibhavi cse25 8 Apr 2026 06:18 am default 63% Present
MJGF 565 8 Apr 2026 06:16 am default 85% Present
MJGF 565 5 Mar 2026 08:49 am default 81% Present
vaibhavi cse25 4 Mar 2026 05:58 am default 72% Present
monica cse30 4 Mar 2026 05:58 am default 78% Present
monica cse30 4 Mar 2026 03:37 am default 73% Present

4. Publication

4th International Conference on Data Analytics and Insights : Submission (317) has been created. Inbox

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Hello,

The following submission has been created.

Track Name: Wireless and Sensor Communication Systems

Paper ID: 317

Paper Title: Automatic student Attendance monitoring and analytics system

Abstract:
The rapid evolution of digital technologies has
significantly transformed traditional academic processes, including
attendance management in educational institutions. Conventional
methods such as manual roll calls and paper-based registers are not

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